

GREENVILLE DOWNTOWN, SC, USA

WMO: 723119

Lat: 34.846N

Lon: 82.346W

Elev: 319

StdP: 97.55

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 13886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-2.9	-17.1	0.9	-0.7	-14.8	1.1	0.8	9.9	6.7	8.9	8.7	2.7	10	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	34.5	22.8	33.3	22.7	32.3	22.5	24.7	30.8	24.2	29.8	23.8	29.3	3.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.4	26.4	22.6	18.0	26.2	22.3	17.6	26.1	76.2	31.1	74.3	30.3	72.5	29.3	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.5	7.6	6.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-8.7	36.3	2.1	2.2	-10.2	37.8	-11.4	39.1	-12.6	40.3	-14.1	41.9		
			-10.4	25.4	1.9	0.5	-11.8	25.8	-12.9	26.0	-14.0	26.3	-15.4	26.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	6.2	8.2	12.4	17.1	21.4	25.3	26.7	26.3	23.3	17.4	11.8	7.9
	DBStd	8.26	5.22	4.82	4.87	4.13	3.46	2.39	2.15	2.18	3.19	4.12	4.34	4.59
	HDD10.0	392	142	84	32	3	0	0	0	0	0	2	31	97
	HDD18.3	1534	378	284	192	71	15	1	0	0	4	67	199	324
	CDD10.0	2957	23	34	106	214	352	459	517	507	398	230	84	32
	CDD18.3	1057	0	1	7	33	109	209	259	248	152	36	3	1
	CDH23.3	9433	1	3	57	294	868	1899	2547	2308	1187	251	16	1
	CDH26.7	3669	0	0	8	59	263	776	1120	964	427	52	1	0
Wind	WSAvg	2.9	3.1	3.2	3.4	3.4	3.1	2.8	2.7	2.5	2.9	2.8	2.7	2.8
Precipitation	PrecAvg	1282	106	99	121	100	84	106	124	125	111	83	89	123
	PrecMax	1695	191	177	226	238	194	242	298	395	341	213	218	250
	PrecMin	865	41	44	40	40	28	8	59	19	6	0	20	59
	PrecStd	238	41	31	47	45	40	62	61	77	67	57	42	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	23.5	27.9	30.3	33.2	35.2	36.1	36.3	34.2	30.7	25.8	22.1
		MCWB	14.0	16.2	16.3	18.5	21.1	22.4	23.2	22.5	21.9	21.9	17.3	15.9
	2%	DB	18.7	20.9	24.8	28.3	31.1	33.7	34.4	34.0	32.4	27.8	22.8	19.6
		MCWB	13.7	14.0	15.1	18.0	20.4	22.3	23.3	22.7	21.4	19.3	15.6	15.2
	5%	DB	16.8	18.6	22.7	26.8	29.5	32.4	33.1	32.7	31.0	26.1	21.1	17.5
		MCWB	12.2	12.9	14.3	17.2	20.1	22.0	23.3	22.9	21.3	18.3	15.0	13.2
	10%	DB	14.2	16.4	20.7	24.9	27.8	31.0	32.0	31.4	29.2	24.2	19.0	15.3
		MCWB	10.5	11.0	13.4	15.9	19.4	21.7	23.0	22.7	20.8	17.9	13.9	12.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	18.4	18.2	21.2	23.4	24.6	25.5	25.3	24.2	23.5	20.1	18.6
		MCDB	19.4	21.8	23.2	27.2	28.1	31.3	32.2	31.4	29.4	27.3	21.7	20.2
	2%	WB	15.3	16.5	17.2	19.9	22.3	23.9	24.8	24.7	23.6	21.9	18.2	16.5
		MCDB	17.6	18.9	21.8	25.4	27.3	30.3	31.2	30.3	28.4	25.6	20.4	18.8
	5%	WB	13.2	14.4	16.1	18.9	21.5	23.3	24.3	24.1	22.9	20.3	16.9	14.6
		MCDB	15.4	17.2	20.7	23.8	26.7	29.4	30.2	29.4	27.3	24.0	19.3	16.6
	10%	WB	11.3	12.0	14.8	17.7	20.7	22.7	23.9	23.7	22.2	19.1	15.2	12.3
		MCDB	13.4	15.0	19.0	22.4	26.1	28.5	29.6	28.8	26.5	22.6	18.3	14.4
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.3	11.1	11.5	10.5	9.9	9.6	9.3	9.4	10.4	10.8	9.4
		MCDBR	12.1	12.6	13.2	13.4	12.2	11.6	11.0	11.0	11.3	12.0	12.6	11.1
		MCWBR	7.8	7.7	6.6	5.6	4.5	3.7	3.3	3.3	3.5	5.2	6.8	7.2
	5% WB	MCDBR	9.6	10.5	11.2	11.3	10.1	9.8	9.7	9.2	8.8	9.0	9.1	8.6
		MCWBR	7.7	8.1	6.6	5.6	4.3	3.6	3.4	3.3	3.3	4.9	6.7	7.0
Clear-Sky Solar Irradiance	taub	0.309	0.319	0.353	0.392	0.450	0.476	0.521	0.510	0.414	0.377	0.332	0.306	
	taud	2.512	2.483	2.415	2.305	2.175	2.139	2.038	2.053	2.305	2.389	2.474	2.549	
	Ebn at Noon	901	929	919	894	842	816	777	777	844	849	862	883	
	Edh at Noon	82	94	110	129	149	154	170	164	122	103	85	75	
All-Sky Solar Radiation	RadAvg	2.67	3.30	4.37	5.44	6.01	6.34	5.94	5.49	4.72	3.93	2.96	2.29	
	RadStd	0.15	0.37	0.29	0.40	0.46	0.45	0.39	0.26	0.42	0.44	0.27	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+293	+191
(47)	Regional (49 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+115

Nomenclature: See separate page